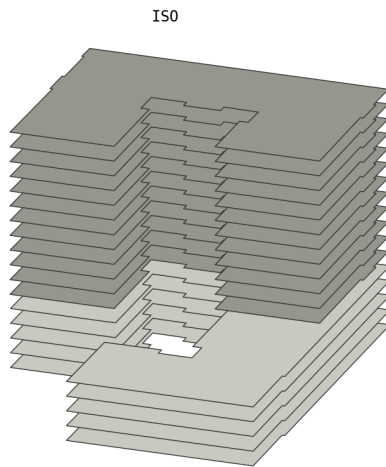


## 1 WIND

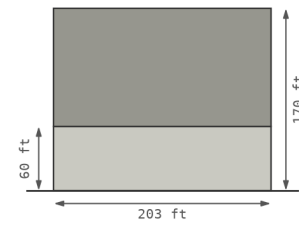
### 1.1 GEOMETRY



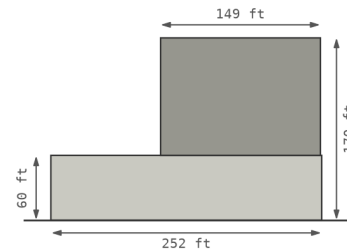
### 115 GILES WIND GEOMETRY - ENVELOPE PROJECTIONS

story height assumed 10 ft - low roof 60 ft, main roof 170 ft

PLAN E-W / VIEW FROM SOUTH



PLAN N-S / VIEW FROM EAST



### 1.2 VELOCITY PRESSURE

$$q_z = 0.00256 K_z K_{zt} K_e K_d V^2 = 0.00256(1.09)(1.0)(1.0)(0.85)(115)^2 = 31.4 \text{ psf} \quad (1)$$

$K_z$  exposure coefficient, Exp B at  $z = h = 170$  ft (ASCE 7-22)

$K_{zt}$  topographic factor

$K_e$  ground elevation factor

$K_d$  directionality factor

$V$  basic wind speed, 115 mph, Risk Cat II, Jersey City NJ

Adopt  $q_z = 35$  psf uniform over full height ( $\approx 11\%$  over  $q_h$ ).

### 1.3 NET PRESSURE (WINDWARD + LEEWARD)

$$p = q_z G (C_{p,ww} + |C_{p,lw}|) = 35(0.85)(0.8 + 0.5) = 38.7 \text{ psf} \quad (2)$$

$G$  gust-effect factor

$C_{p,ww}$  windward wall pressure coefficient, +0.8

$C_{p,lw}$  leeward wall pressure coefficient,  $-0.5$  ( $L/B \leq 1$ ; N-S wind  $L/B \approx 1.25 \rightarrow -0.45$ , use  $-0.5$ )

Adopt  $p = 40$  psf net on projected area, both directions.

### 1.4 BASE SHEAR

$$V_x = p A_x = 40 (252 \cdot 60 + 149 \cdot 110) / 1000 = \boxed{1,260 \text{ k}} \quad (3)$$

$$V_y = p A_y = 40 (203 \cdot 170) / 1000 = \boxed{1,380 \text{ k}} \quad (4)$$

$A_x$  projected area for wind along X (E-W), N-S elevation profile

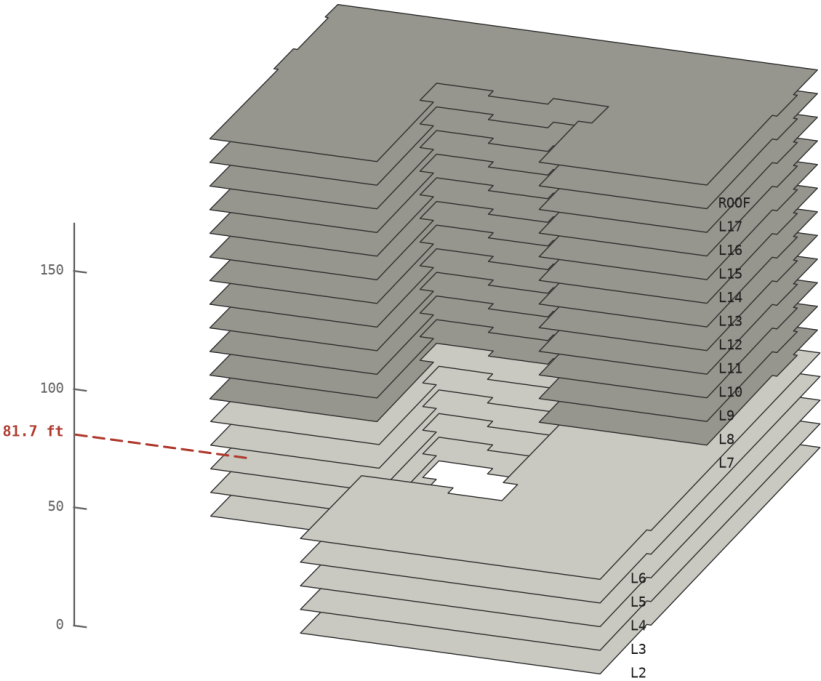
$A_y$  projected area for wind along Y (N-S), E-W elevation profile

2 EQ

2.1 GEOMETRY

|         |                                         |              |
|---------|-----------------------------------------|--------------|
| LOWER   | 35,765 sf x 5 (L2-L6)                   | = 178,823 sf |
| UPPER   | 23,141 sf x 12 (L7-R00F)                | = 277,691 sf |
| TOTAL   |                                         | = 456,514 sf |
| MEAN HT | $\text{sum}(A \cdot z) / \text{sum}(A)$ | = 81.7 ft    |

**115 GILES**  
**CRUDE SLAB MASSING**  
story height assumed 10 ft



2.2 LOADS

$$w = w_{slab} + w_{SDL} = 100 + 35 = 135 \text{ psf} \tag{5}$$

$w_{slab}$  8 in slab self-weight, 100 psf, all floors  
 $w_{SDL}$  superimposed dead load, 35 psf, all floors

$$W = w \Sigma A = 0.135(456,514) = 61,629 \text{ k} \tag{6}$$

$W$  effective seismic weight  
 $\Sigma A$  total slab area above base, per 2.1

### 2.3 HAZARD / SITE

Case study: ASCE 7-16 site-coefficient chain vs ASCE 7-22 direct multi-period, both Site Class D assumed (no geotech).

#### ASCE 7-16

$$\begin{aligned} S_{MS} &= F_a S_S = 1.58(0.28) = 0.44 \\ S_{M1} &= F_v S_1 = 2.40(0.051) = 0.12 \\ S_{DS} &= \frac{2}{3} S_{MS} = 0.30 \\ S_{D1} &= \frac{2}{3} S_{M1} = 0.082 \end{aligned}$$

$S_S, S_1$  mapped  $MCE_R$  accelerations, 0.28 / 0.051  
 $F_a, F_v$  site coefficients (7-16 Tables 11.4-1/2)  
 $S_{MS}, S_{M1}$  site-adjusted  $MCE_R$ ; 7-22 values direct from ASCE 7 Hazard Tool  
 $S_{DS}, S_{D1}$  design accelerations

#### ASCE 7-22

$$\begin{aligned} S_{MS} &= 0.30 \text{ (tool, direct)} \\ S_{M1} &= 0.10 \text{ (tool, direct)} \\ S_{DS} &= \frac{2}{3} S_{MS} = 0.20 \\ S_{D1} &= \frac{2}{3} S_{M1} = 0.067 \end{aligned}$$

Risk Cat II,  $I_e = 1.0$ . SDC B both cases.

### 2.4 SYSTEM / PERIOD

Ordinary reinforced concrete shear walls, bearing wall system,  $R = 4$ .

$$T_a = C_t h_n^x = 0.02(170)^{0.75} = 0.94 \text{ s} \quad (7)$$

$R$  response modification coefficient  
 $C_t, x$  approximate-period coefficient / exponent, all other systems  
 $h_n$  structural height above base, 170 ft

### 2.5 BASE SHEAR (ELF)

$$C_s = \frac{S_{DS}}{R/I_e} \quad C_{s,max} = \frac{S_{D1}}{T(R/I_e)} \quad C_{s,min} = 0.044 S_{DS} I_e \geq 0.01 \quad V = C_s W \quad (8)$$

$C_s$  seismic response coefficient, bounded by  $C_{s,max}$  /  $C_{s,min}$   
 $T$  period used in direction considered,  $T = T_a = 0.94 \text{ s}$

#### ASCE 7-16

$$\begin{aligned} C_s &= 0.30/4 = 0.075 \\ C_{s,max} &= \frac{0.082}{0.94(4)} = 0.0218 \leftarrow \\ C_{s,min} &= 0.044(0.30) = 0.0132 \\ V &= 0.0218(61,629) = \boxed{1,344 \text{ k}} \end{aligned}$$

#### ASCE 7-22

$$\begin{aligned} C_s &= 0.20/4 = 0.050 \\ C_{s,max} &= \frac{0.067}{0.94(4)} = 0.0178 \leftarrow \\ C_{s,min} &= 0.01 \text{ (floor)} \\ V &= 0.0178(61,629) = \boxed{1,097 \text{ k}} \end{aligned}$$

7-16 governs, +23%.  $C_{s,max}$  controls both cases.